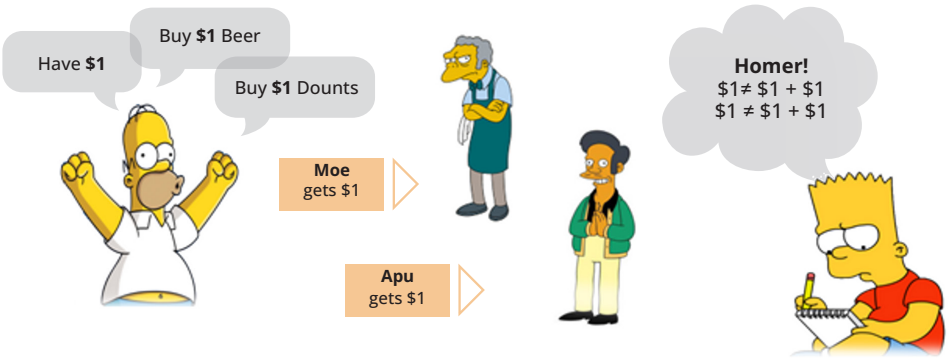
5.1.11 How does the Blockchain keep track of who spends which bitcoin?

You know how you can't spend the same dollar bill twice? ***Bitcoin*** similarly works to make sure that you can't spend the same digital coin twice.

You see, before ***Bitcoin***, it was possible to send transactions across a network of computers, but there was a problem: people could send conflicting transactions, like trying to spend the same coin twice. This is called "**double-spending**".



Bitcoin solves this problem by making all the computers on the network work together. When a new transaction is sent, it's sent to all the computers, and they keep it in memory before writing it to a permanent file (the *blockchain*).

This process is called "**mining**", and it makes sure that no double-spending transactions ever get written to **the file**. It's like a big competition that nobody can cheat, so your ***bitcoin*** are always safe.

This is how *Bitcoin* reaches consensus, all without a single punch being thrown!

At regular intervals, one of the computers gets to add all the transactions it has in memory to the file. Then it shares the updated file with all the other computers on the network. All the computers agree on which transactions are valid, and which aren't, and they remove any conflicting ones from their memory.

5.1.12 How do new bitcoin enter the network?

To pay or **reward miners** for their hard job, every time they add a new block to the blockchain, they get compensated with newly minted ***bitcoin***. Currently miners get paid 6.25 BTC for every block they mine.

5.1.13 What is a bitcoin transaction?

A ***bitcoin*** **transaction** is a transfer of ownership of existing ***bitcoin*** units to a new owner. But instead of transferring actual coins, what happens is that all the nodes in the network update their local copy of the public ledger to reflect the change in ownership. (Think back to the Rai stones! This is just a more advanced version, where the ledger is externalized, rather than memorized, for all the public to review and see).